Content

Learning objectives and content

Objectives
At the end of the course, students should be able to:

- Understand the behaviour of all stochastic models that have been discussed in the course.
- Identify the appropriate simulation model to describe the practical situations involving randomness.
- Design and implement a simulation model, for all discussed models, but also for complicated models based on real-life situations, in a programming language of their choice.
- Work efficiently in a team, when developing and implementing a simulation model.
- Analyse data taken from real life in order to use (abstractions of) this data in simulation experiments.
- Analyse (and, if necessary, improve) the accuracy of the simulation outcomes
- Interpret the results of simulation experiments.
- Write a comprehensive report in which all aspects of the development and implementation of the simulation model are described, as well as a summary of the results, and
- conclusions and recommendations for the problem owner, based on these results

Content
Please note that some of the lectures require compulsory attendance.

Many real-life processes are too difficult or too complex to analyse in an exact, theoretical way. Throughout this course we discuss a variety of real-life processes that exhibit a stochastic behaviour, ranging from financial models, manufacturing networks, to models for particle systems in chemistry or physics. The course is very practical, in the sense that emphasis will be placed on developing simulation models for these processes. Many of the lectures will be interactive, with the lecturer and the students writing a simulation together, step by step. The main goal of the course is that the students not only learn and understand the required simulation techniques, but are also forced to actually write the simulation.

The practical aspect is also reflected in the way that the assessment takes place: students, working together in groups of two, will be given several assignments. Each assignment addresses a real-life problem. It is not only important that the students are capable of finding a suitable model for the problem, and writing a program that simulates this model, but it is also relevant that the student is able to interpret the outcomes, and use the simulation as a decision support tool to formulate recommendations on how the process might be improved.

Several programming techniques will be discussed, allowing the student to choose whichever suits him best. Although the programming language Python is used during the lectures, the students are free to choose whichever programming language they prefer for their assignments. We also provide sample code in Java and R. The course is aimed at experienced programmers, familiar with concepts like object oriented programming. Students who have followed the course Programming and Modelling (2WH20), or a similar course with respect to the programming topics, should have sufficient preknowledge.

Please note that there is an atypical form of assessment:

- 50% of the grade is determined by a number of assignments (usually 2) that are made in groups of multiple students (also usually 2). Please note that the number of assignments and number of students per group will not be determined until the start of the course.

- 50% of the final grade is determined by an individual test. This can either be a laptop exam, with programming exercises, or a written exam with questions aimed at assessing the student's knowledge about (i) the topics discussed in the course, and/or (ii) implementation details of the group assignments, or a mixture of both. For this last part, a minimum grade of 5 is required. Since there is no such restriction for the first two assignments, there is no possibility to improve their grade. Students will be given a resit opportunity for the last part.

Topics:

- Random number generators

- Generating random variables

- Monte-Carlo simulation

- Discrete-event simulation of several stochastic processes and queueing networks, like:

- Random walks, birth-and-death processes
- Epidemic models
- Financial risk models
- Interacting particle systems
- Communication networks
- Manufacturing systems and networks
- Input and output analysis (distribution fitting, warmup-interval, confidence intervals)

Learning Type
Challenge based

Real life problems (with stakeholders)

Prototyping/modelling

2026

Stochastic Simulation and Modeling (2MBS40)

Practical information

Course module
2MBS40

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Deepening

Language of instruction
English

Offered by
Mathematics and Computer Science - Mathematics domain

Short promotional description of the course
The course discusses techniques to write computer simulations for a variety of real-life processes that exhibit a stochastic behaviour, ranging from epidemic models, financial models, manufacturing networks, to models for interacting particle systems in physics. The course is very practical, and mainly focusses on the modeling aspect, and on developing the simulation. Many of the lectures will be interactive, with the lecturer and the students writing a simulation together, step by step. This course is aimed at students who like to program and extend their programming skills and learn techniques to implement realistic discrete-event systems with stochastic characteristics.

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3
  Timeslot(s)
  A - Mo 1-4, We 9-10, Th 5-8

  Enrolment period
  15 November 2026 until 3 January 2027 23:59

  Unenrolment
  until 1 February 2027 23:59

Entrance requirements

Entrance requirements course enrolment
You must meet the following requirements

- Completed none of the course modules listed below
  - Stochastic simulation (2WB50)

Instructional modes

Instructional modes

- Instruction
  Remark
- Tutorial
  Remark

Tests

Tests

- Assignments
  Interim examination

  Test weight
  50

Minimum grade

- Written examination
  Final examination

  Test weight
  50

  Minimum grade
  5.0

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignments

- Block 3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

Written examination

- Block 3
  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 21 March 2027

  Test date
  12 April 2027

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  28 June 2027

Lecturer(s)

Contactperson for the course

- dr.ir. M.A.A. Boon

Lecturer

- dr.ir. J. Sanders

Responsible lecturer

- dr.ir. M.A.A. Boon

Subject matter expert

- dr.ir. M.A.A. Boon

Materials

Do you want to know more?

dr.ir. M.A.A. Boon can tell you more about this course.
Send an e-mail to m.a.a.boon@tue.nl